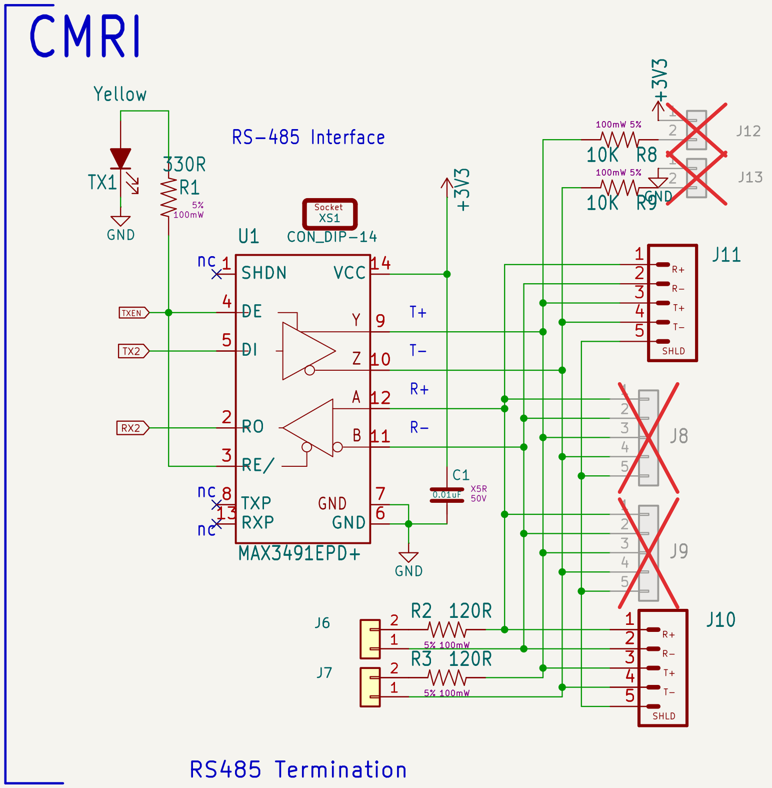


RS-422 uses separate transmit and receive wire pairs for full-duplex communication.
RS-485 typically uses a single shared pair for half-duplex, multi-point networking.

Since CRMI's wire protocol will only USE the bus in a half duplex manner, a 2-pair RS422 bus driver can be configured into a 1-pair RS485 scheme by connecting the two pairs: T+ to R+ and T- to R-.

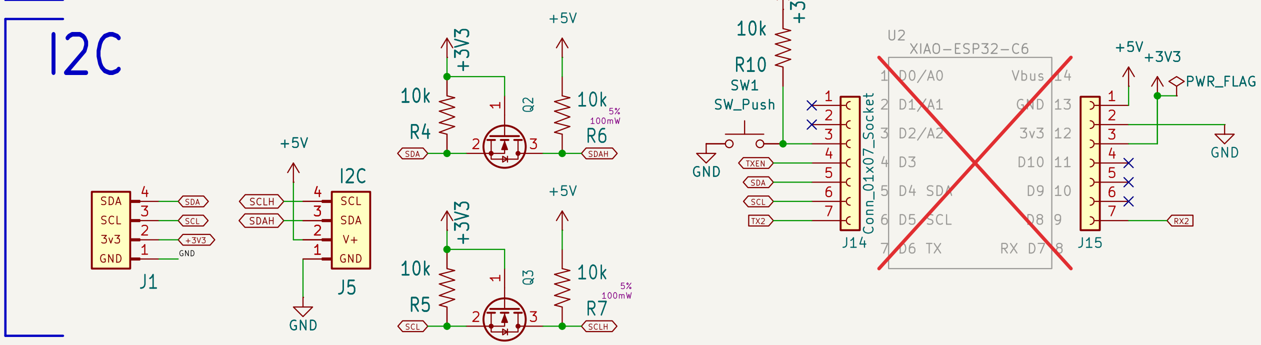
In addition, on the software side, the CMRI protocol library must actively control the TXenable/RXenable pins on the bus interface IC when used on a 1-pair bus

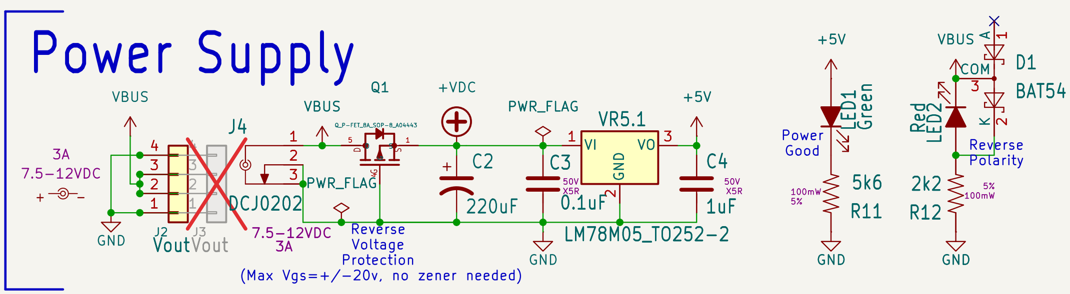
Bias is used in a 1-pair half duplex RS485 wiring configuration.
Biasing A+ to v+ and B- to GND ensures no RX "Null char noise" gets injected if a bus driver is misconfigured to have RE tied to gnd
Only configure a single node to add bias if it is needed.

Termination is a noise prevention requirement that prevents "ringing" when transmissions hit the end of a transmission line.

The Shield wire is used ONLY in the cables between nodes to reduce the impact of electrical noise on the bus. The shield should be connected to earth ground at ONE END Only. Do not connect the shield to power supply GND.

Each pair used in the bus should be terminated at either end.
A 4-wire/2-pair RS422 bus uses both RX and TX terminations, one for each pair.
A 2-wire/1-pair RS485 bus only needs one termination on the single A+/B- pair.





open hardware

SPCoast
Fab Info
cpNode Xiao
Board Boilerplate

Add CAN TERM. Bias, TXEN/!RXEN, alt RS485 pins SPCoast

Sheet: /
File: cpNode-Xiao.kicad_sch

Title: cpNode Xiao John Plocher

Size: USLetter	Date: 2026.09	Rev: 3.1
KiCad E.D.A. 10.0.1	Id: 1/1	active